

Assignment 3

CSE471

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Section : 9

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Ans. to the Q.No.1

According to the question's scenario, an object oriented data base ~~for~~ would be most suitable here as it naturally supports complex objects, object identity, inheritance, encapsulation, object oriented application code. An object oriented database stores information as objects containing both attribute (data) and behaviors (method). As the application here needs to preserve complex relationships, inheritance and object identity, object oriented database is perfect here. It reduces mismatch between application objects and stored data. Objects created in the application can be stored in a form closely resembling original object oriented representation. That is the tight coupling between application logic and data model.

Ans. to the Q.No.2

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So,

Fields	Arg. Size
Order Number	$2+5 = 7$
Date	$2+5 = 7$
Customer ID	$2+5 = 7$
Last Name	$9+5 = 14$
First Name	$9+5 = 14$
State	$4+5 = 9$
Amount	$8+5 = 13$
Tax Rate	$1+5 = 6$

$$\textcircled{b} \text{ Record size} = 7+7+7+14+14+9+13+5 \\ = 76 \text{ characters}$$

$$\textcircled{c} \text{ Total Record size} = \text{Record size} + (\text{Record size} \times \text{overhead}) \\ [\text{overhead} = 0.35] \\ = 76 + (76 \times 0.35) \\ = 76 + 26.6 \\ = 102.6 \text{ characters}$$

$$\textcircled{d} \text{ Initial table size} = (\sum id) \times 1200 \\ = (2+2+2+9+9+4+8+0) \times 1200 \\ = 36 \times 1200 \\ = 43200 \text{ records}$$

$$\therefore \text{Initial table volume} = \text{Initial table size} \times \text{Total Record Size} \\ = 43200 \times 102.6 \\ = 4432320 \text{ characters.}$$

$$\textcircled{e} \text{ After 1095 days, } = \frac{1095}{365} = 3 \text{ years} \quad \left| \begin{array}{l} \text{growth rate} \\ \text{(quarterly)} \\ = 1500 \end{array} \right.$$

Since, quarterly growth = $3 \times 4 = 12$ quarters

$$\therefore \text{Final table volume} = \left\{ \text{Initial table volume} + (\text{time} \times \text{growth}) \right\} \times 102.6 \quad \leftarrow \text{total record size} \\ = 4432320 + (12 \times 1500) \times 102.6 \\ = 6,279,120 \text{ characters (Ans)}$$

Ans. to the Q.No. 3

Cohesion: How good the lines of code written within each module relate to each other.

① Functional Cohesion: Every operation in the module, contributes to the ~~an~~ one specific function.

```
def calculate_fine(days, rate):  
    return days * rate
```

calculate_fine

② Sequential Cohesion: The output of one operation becomes the input of the next operation.

```
data = validate_user(data)  
account = create_account(data)  
setup_profile(account)
```

③ Communicational Cohesion: Several operations are grouped together as they work with the same data.

```
def process_book(book):  
    display_details(book)  
    check_availability(book)  
    show_status(book)
```

process_book

④ Procedural Cohesion: Tasks are done in a sequence/order.

```
def logout_user():  
    log_event()  
    end_session()  
    redirect()
```

logout_user

④ Temporal Cohesion: Tasks are executed at the same time.

```
def start-system():  
    load-config()  
    connect-database()  
    start-logging()
```

start-system

Ans. to the Q No. 4

